Lab 4 — React Routing and State Management

Aim

Build a multi-page single-page application (SPA).

Input / Source Code

App.jsx import {

```
 BrowserRouter,
 Routes,
 Route,
 Link
} from "react-router-dom";
import { createContext, useContext, useState } from "react";
const CartContext = createContext();
const products = [
 { id:1,
   name: "Laptop",
price: 55000,
description: "High-performance laptop for students."
 },
{
   id: 2,
   name: "Mobile Phone",
   price: 25000,
   description: "Powerful smartphone with modern features."
},
{
Id: 3,
Name: “Headphones”,
price: 3000,
 description: "Wireless headphones with clear sound."
 }
];
```

```
function Home() {
  return (
    <div>
      <h2>Home Page</h2>
      <p>Welcome to our React Store.</p>
    </div>
  );
}
function Products() {
return (
    <div>
      <h2>Products</h2>
      {products.map((product) => (
        <div key={product.id}>
          <h3>{product.name}</h3>
          <p>Price: ₹{product.price}</p>
          <Link to={`/products/${product.id}`}>
            View Details
          </Link>
        </div>
      ))}
    </div>
  );
}
function ProductDetails() {
const { id } = useParams();
  const { addToCart } = useContext(CartContext);
  const product = products.find(
    (item) => item.id === Number(id)
  );
  if (!product) {
```

```
    return <h2>404 - Product Not Found</h2>;
  }
  return (
    <div>
      <h2>Product Details</h2>
      <h3>{product.name}</h3>
      <p>Product ID: {product.id}</p>
      <p>Price: ₹{product.price}</p>
      <p>{product.description}</p>
      <button onClick={() => addToCart(product)}>
        Add to Cart
      </button>
    </div>
  );
}
function NotFound() {   return <h2>404 -
Page Not Found</h2>;
}
function App() {   const [cart,
setCart] = useState([]);
  const addToCart = (item) => {
setCart([...cart, item]);
  };
  return (
    <CartContext.Provider value={{ cart, addToCart }}>
      <BrowserRouter>
        <h1>React Store</h1>
        <nav>
          <Link to="/">Home</Link> |{" "}
          <Link to="/products">Products</Link>
        </nav>
```

```
        <p>Cart Items: {cart.length}</p>
       <Routes>
        <Route path="/" element={<Home />}
 />
        <Route
path="/products"
element={<Products />}
/>
        <Route
path="/products/:id"
element={<ProductDetails />}
        />
        <Route
 path="*"
element={<NotFound />}
        />
       </Routes>
     </BrowserRouter>
   </CartContext.Provider>
  );
}
export default App;
```

Note: ProductDetails mein useParams() import bhi add karna hai:

```
import { useParams } from "react-router-dom";
```

Output

1. Home Page

React Store

Home | Products

Cart Items: 0

Home Page

Welcome to our React Store.

2. Products Page

React Store

Home | Products

Cart Items: 0

Products

Laptop

Price: ₹55000

[View Details]

Mobile Phone

Price: ₹25000

[View Details]

Headphones

Price: ₹3000

[View Details]

3. Product Details — Route ID 1

URL:

/products/1

Output:

React Store

Home | Products

Cart Items: 0

Product Details

Laptop

Product ID: 1

Price: ₹55000

High-performance laptop for students.

[Add to Cart]

4. Product Details — Route ID 2

URL:

/products/2

Output:

Product Details

Mobile Phone

Product ID: 2

Price: ₹25000

Powerful smartphone with modern features.

[Add to Cart]

5. Product Details — Route ID 3

URL:

/products/3

Output:

Product Details

Headphones

Product ID: 3

Price: ₹3000

Wireless headphones with clear sound.

[Add to Cart]

6. Cart State

After clicking **Add to Cart**:

Cart Items: 1

7. Invalid Product ID

URL:

/products/10

Output:

404 - Product Not Found

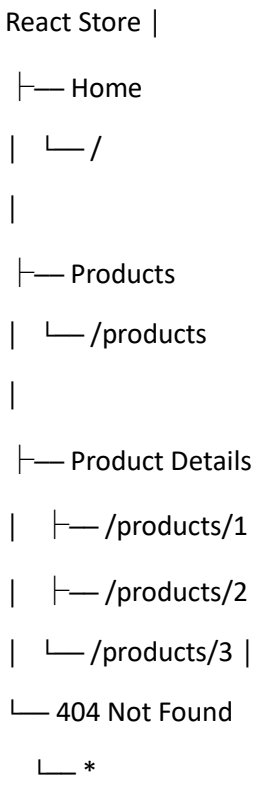
8. Invalid Page URL:

/about

Output:

404 - Page Not Found

Routing Structure


React Store |
├— Home
| └— /
|
├— Products
| └— /products
|
├— Product Details
| ├— /products/1
| ├— /products/2
| └— /products/3 |
└— 404 Not Found
└— *